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M. B. E. S's COLLEGE OF ENGINEERING, AMBAJOGAI

Project Title:

Women Safety App & Security with SMS Alert based Android app

Project Guide : Prof. V. V. Gore

Department :- Third Year Computer Science and Engineering

YEAR 2025 – 2026

INTRODUCTION :

- Women safety is an important concern in today’s society
- In emergency situations, quick communication and location tracking are very important.
- Our Women Safety App is developed to provide fast help and emergency support.
- The app sends SOS alerts with current location to emergency contacts.
- It also includes a feature to send updated location SMS every 5 minutes for continuous tracking.
- This project helps improve personal safety, security, and quick response during danger situations.

PROBLEM STATEMENT

- Women may face unsafe situations while traveling or alone.
- It becomes difficult to quickly inform family or emergency contacts during danger.
- Immediate location sharing is important for fast help and safety.

Existing Issues or Gaps

- Many existing systems do not provide continuous live location updates.
- Some applications are complex and not easy to use in emergencies.
- Lack of quick SOS communication can delay emergency response.

OBJECTIVES

Main Objectives

- To develop a mobile application for women safety and emergency support.
- To provide quick SOS alerts with current location sharing.
- To improve user safety through simple and fast communication features.

Measurable Goal

- Send emergency SMS to saved contacts instantly.
- Share current location using GPS tracking.
- Provide automatic location updates every 5 minutes.
- Allow users to manage emergency contacts easily.
- Create a simple and user-friendly interface.

LITERATURE REVIEW

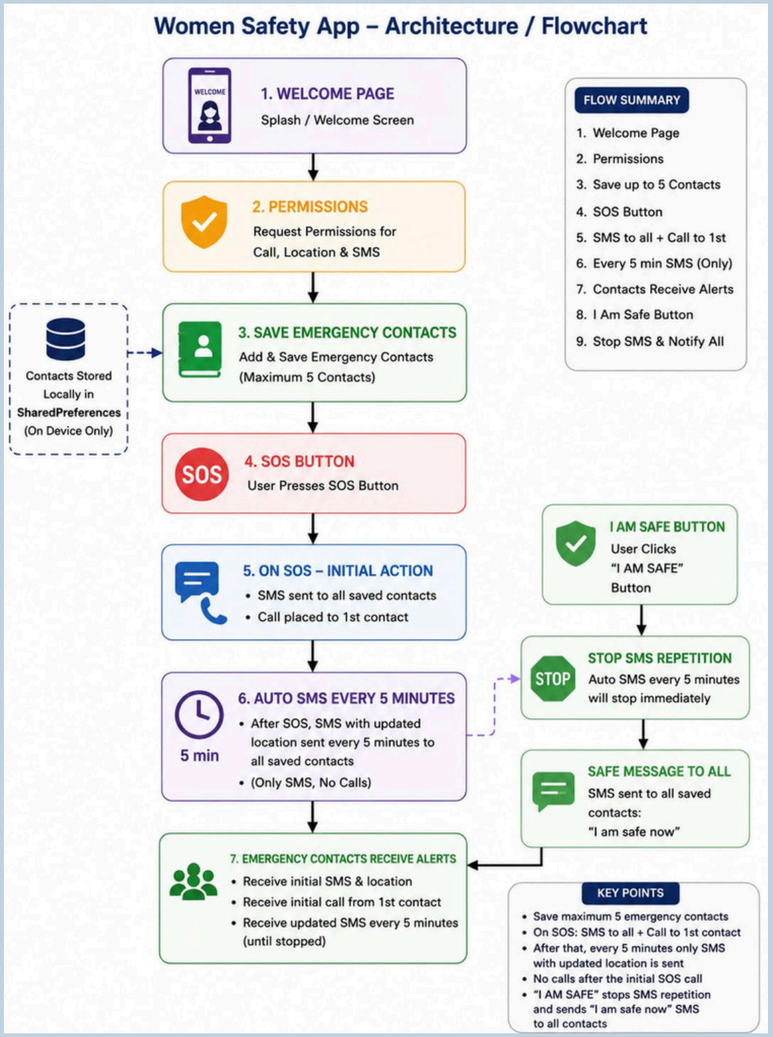
Existing Systems

Existing women safety applications provide features like SOS alerts, emergency calling, and live location sharing. Some apps also use GPS tracking and SMS services to contact emergency numbers during unsafe situations. These systems help users communicate quickly in emergencies.

Gaps in Existing Systems

Many existing apps only send location once and do not provide continuous tracking. Some applications require internet connection all the time, which may not work properly in all situations. Complex interfaces and delayed emergency response are also common limitations in current systems.

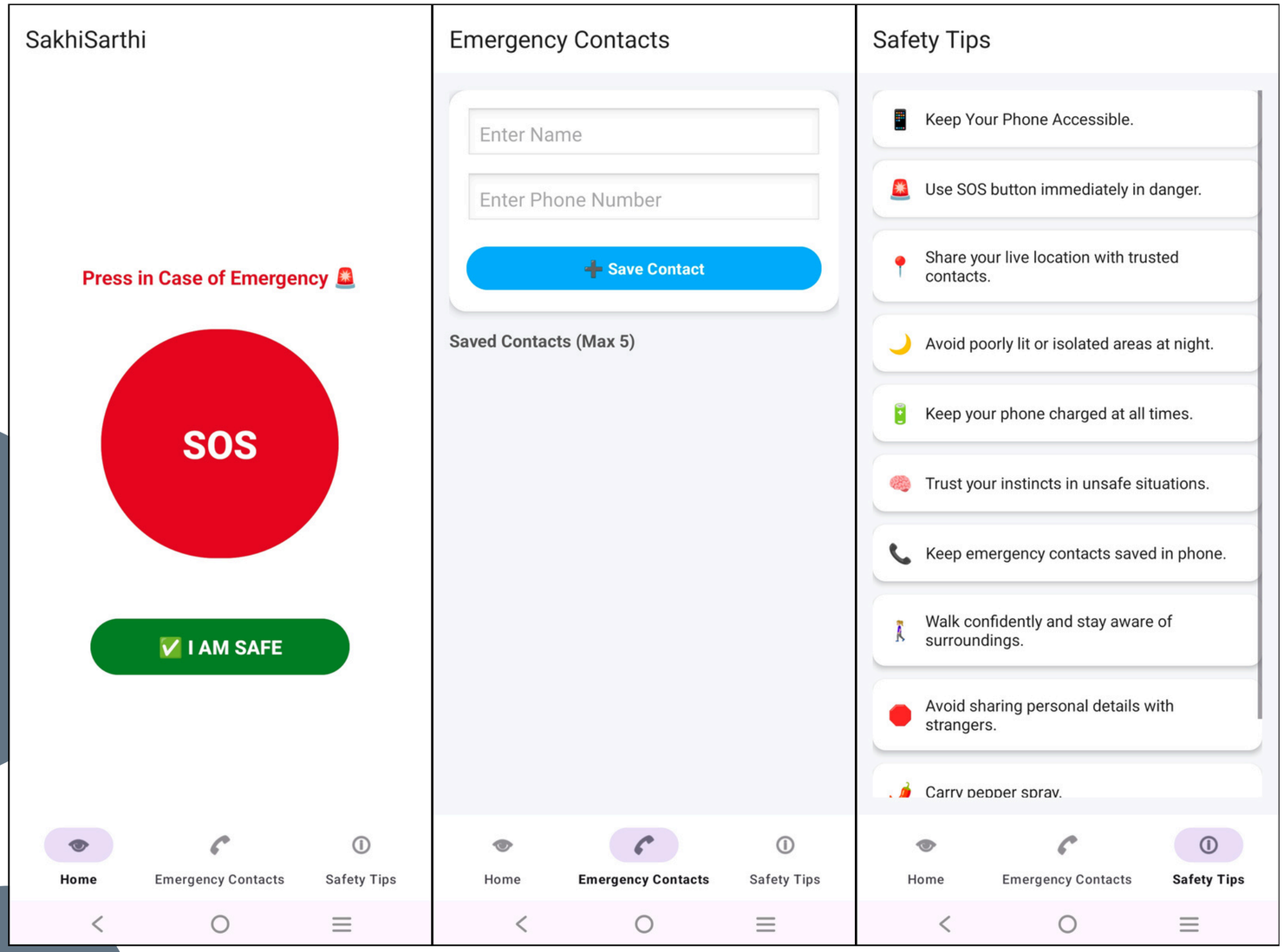
PROPOSED METHODOLOGY



Technologies Used :

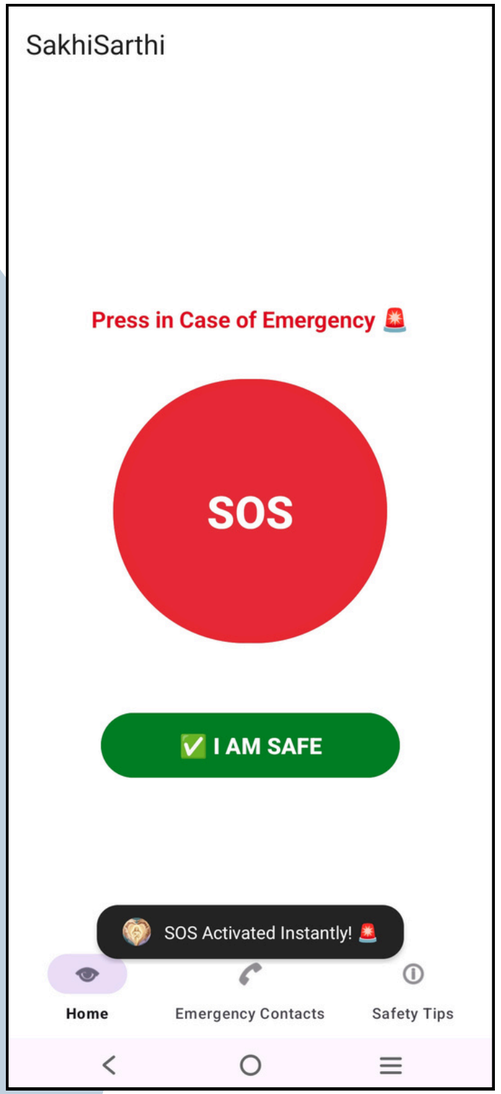
- Android Studio
- Java (Android Development Language)
- XML (User Interface Design)
- GPS / Location Services (for live location tracking)
- SMS Manager (for sending emergency messages)
- Telephony & Call Permissions (for emergency calling feature)
- SharedPreferences (for storing contacts locally on device)

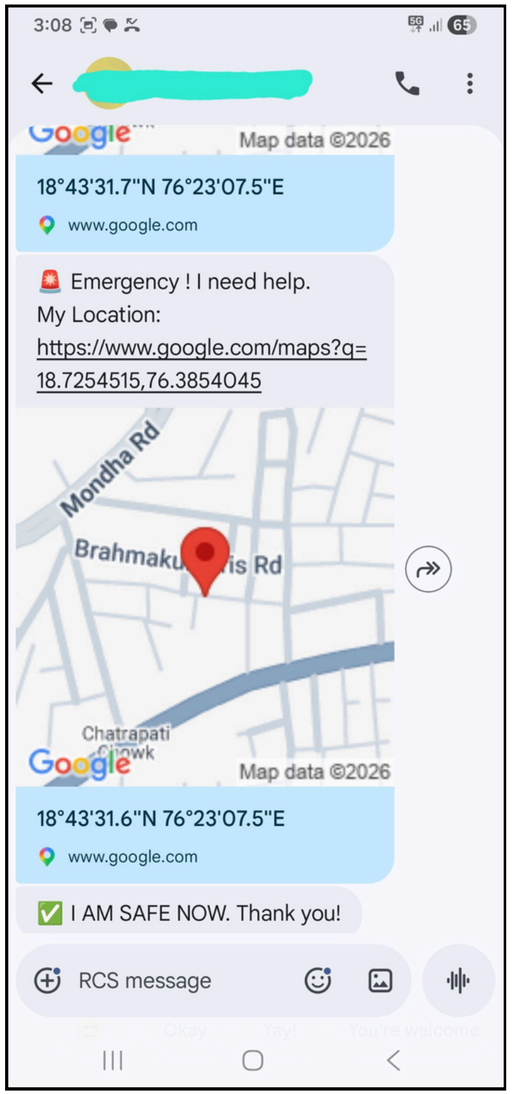
IMPLEMENTATION

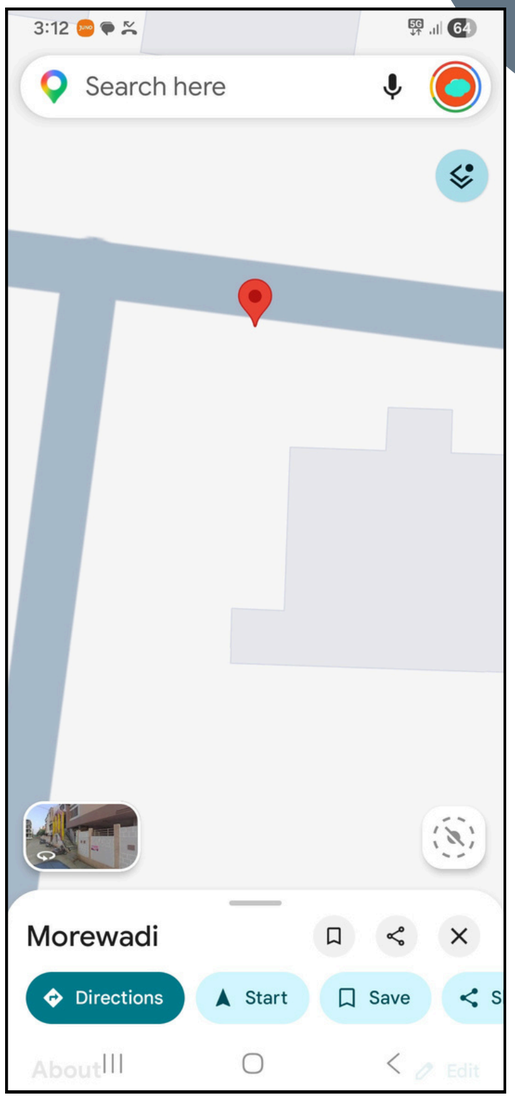


- User opens the app and grants permissions (Location, SMS, Call).
- Saves up to 5 emergency contacts locally.
- On pressing SOS, SMS with current location is sent to all contacts and call is made to the first contact.
- After SOS, SMS with updated location is sent every 5 minutes automatically.
- This continues until the user presses “I am Safe” on the homepage.
- On clicking it, repetition of SMS stops and an “I am safe now” SMS is sent to all contacts.

RESULT & ANALYSIS







CONCLUSION :

The Women Safety Application provides a quick and reliable emergency response system. It ensures user safety by sending current location alerts, making emergency calls, and continuously updating contacts during critical situations. The app is simple to use and helps users get immediate assistance during emergencies.

FUTURE SCOPE:

In the future, the SOS button can be integrated directly into the phone’s lock screen for faster access without opening the app. Additional features like real-time location tracking on maps, voice activation for SOS, and integration with police or emergency services can also be added to enhance safety and response efficiency.

REFERENCES

Books

- Android Programming: The Big Nerd Ranch Guide
- Head First Android Development

Research Papers

- Emergency Alert and Safety Systems using Mobile Applications
- Location-Based Services in Android Applications

Websites

- https://developer.android.com (Android Developers Documentation)
- https://www.geeksforgeeks.org (Android Development Tutorials)
- https://www.stackoverflow.com (Programming Help & Solutions)

Thank You !